



Session 4: Aircraft Performance

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Airbus Flysmart App

A320FAMILY

Airbus Flysmart App

Airbus Takeoff application

The Takeoff module is an OCTOPUS program (except for Wide Body aircraft, for which TLC computation engine is used).

This module aims at computing the takeoff performance data (maximum takeoff weight, takeoff speeds, flexible temperature and the All Engine Climb Gradient (AECG) capability (except for Wide body aircraft for AECG)) according to the aircraft configuration and the external conditions (runway, surrounding obstacles, and weather).

Start the Airbus Takeoff application by tapping the AIB Takeoff icon in the “My flight” page. Start the Airbus Takeoff application by tapping the AIB Takeoff icon in the “My flight” page.



Airbus Flysmart App

Airbus Takeoff application

The main interface of the Takeoff application is displayed:

The interface is composed of the following panels:

1. Airport Data panel: By default, the departure airport set in the Flight Ops status page is selected if applicable to the selected aircraft. The User may choose another airport and select one or several runways to perform a computation.
2. Conditions Data panel: Enables to define the runway conditions and the aircraft configuration.

The screenshot shows the 'My Flight' screen with the 'TAKEOFF' tab selected. The flight information at the top includes 'A9C-AC' and 'A320-214'. The left panel contains input fields for airport data (DBB/BAH, BAHRAIN), runway (RWY), wind (WIND °/kt, (000/0)), OAT (°C), QNH (hPa), RWY COND, A-ICE, TOW (T, 77.0), T.O CG (CG > 27% (STD)), CLB THRUST (CL), CONF (CONF 1+F (STD)), and AIR COND (OFF (STD)). At the bottom of the left panel are buttons for 'MEL 0', 'CDL 0', 'CLEAR', and 'COMPUTE'. The right panel is titled 'TAKEOFF' and displays a large message: 'All fields must be completed to compute your take-off performance data'. The bottom navigation bar includes icons for a calendar, a plane, a person, and a folder.

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AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

Once the airport is selected a list of available RWY's will appear each with its available intersections.



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AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

After the runway selection, the runway characteristics are displayed if user taps on the runway drawing that is displayed on the right hand side of the screen. They consist in TORA / TODA / runway elevation, slope, entry angle, number of obstacles, takeoff shift and number of all engine gradient constraints.

The screenshot shows the 'MODIFY 30R' screen in the Airbus Flysmart app. The screen is divided into several sections:

- Top Bar:** Displays 'My Flight', 'TAKEOFF', 'AFC AC', and 'A320-214'.
- Left Panel:** Contains flight parameters and conditions:
 - 0881/BAH BAHRAIN >
 - RWY 30R >
 - WIND °/kt (280/ 20/ G5)
 - OAT °C 45 (ISA +30)
 - QNH hPa 1013
 - RWY COND Dry >
 - A-ICE Off >
 - TOW T 77.0
 - T.O CG CG > 27% (STD) >
 - CLB THRUST CL
 - CONF CONF 1+F (STD) >
 - AIR COND Off (STD) >
- Right Panel (MODIFY 30R):** Contains runway characteristics and obstacles:
 - REDUCTION FROM RUNWAY START 0 m
 - TORA 3964 m
 - TODA 3964 m
 - ASDA 3964 m
 - WIDTH 45 m
 - OBSTACLES +
 - OBSTACLES FROM RWY START 2
 - MODIFY Entry Angle
 - 30R-A9 No change >
 - 30R No change >
 - 30R-A8 No change >
 - 30R-A6 No change >
 - 30R-A5 No change >
 - Restore Done
- Bottom Bar:** Contains buttons for MEL 0, CDL 0, CLEAR, and COMPUTE.

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Airbus Takeoff application

AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

Reduced Mode

The User can reduce the runway length from the beginning and / or the end of the runway and change the runway width. He can also add an obstacle or modify the entry angle of the main runway or of the runway entries. These modifications should be done in accordance with NOTAMs.

To modify a runway:

- Tap on Modify RWY button below the runway list.
- In the runway list, tap on the Modify RWY button associated to the main runway to modify.



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Airbus Takeoff application

AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

Obstacle Details

For detailed information about the obstacles, tap on the View OBSTACLES button in order to display the obstacles panel. The obstacles panel displays the obstacles information (distances and height) with reference to the threshold of the selected runway.

If a runway entry is selected, the distances and height are in reference to the runway entry threshold.



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AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

Note that the ALL ENGINE GRADIENT field and the Modify All Engine Data button are available for all Airbus aircraft.

THR RED HEIGHT / ACC HEIGHT

The User can modify the Thrust Reduction height or/and the Acceleration height.



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AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

CLB SPD LIM

The User can choose to enable or disable the climb speed limit. If the speed limit is enabled, the user can modify the altitude or the speed of the climb speed limit.

FINAL CLB SPD

The User can modify the final climb speed.



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AIRPORT PANEL: SINGLE RUNWAY COMPUTATION

ALL ENGINE GRADIENT

The User can add gradients by clicking on the cross. The number of constraints is limited to 10 by runway.

A gradient can be defined with:

- A gradient and a target altitude
- Only a target altitude and no gradient value. In this case, the gradient value is set to 3.3%.

If the user taps on existing constraint, he has the possibility to modify this constraint.



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TOW

The User enters here the takeoff weight used for the calculation. This can be either the actual takeoff weight of the aircraft or the structural MTOW if the User wants to calculate the maximum takeoff weight limited by performance.

TOW	77.0 T
T.O CG	CG > 27 % (STD) >
T.O. THRUST TOGA /FLEX (STD)	
CLB THRUST	CL
CONF	CONF 3 >
AIR COND	On (STD) >

T.O CG

The field Takeoff CG is given for some aircraft types that have two or more certified forward CG limits at takeoff. Set the value here according to the aircraft CG at takeoff. Standard setting is CG > 27% STD and CG < 27%.

T.O CG

Cancel

Enter your CG selection.

CG > 27 % (STD)
CG < 27 %

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CLIMB THRUST

The Climb thrust field in the Conditions Data panel is only available when the administrator activates the all engine gradient function, it is set on the CL option.

CONF

The User can select the takeoff configuration.

The possible takeoff configurations are:

CONF 1+F STD, CONF 2, CONF 3 and OPT CONF for Fly-By-Wire aircraft.

When OPT CONF is selected:

If takeoff is limited by performance, the calculation returns automatically the configuration that maximizes the performance-limited MTOW. If takeoff is not limited by performance and flexible takeoff is possible, the calculation returns the configuration that maximizes the flexible temperature (to reduce engine wear). If takeoff is not limited by performance and two different flap configurations provide the same maximum flexible temperature, the calculation returns the configuration that minimizes the V2/VS ratio.

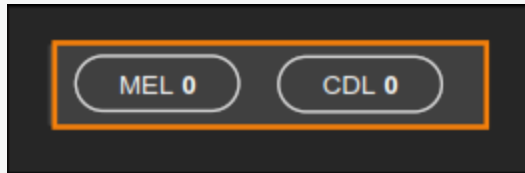
TOW	77.0 T
T.O CG	CG > 27 % (STD) >
T.O. THRUST TOGA /FLEX (STD)	
CLB THRUST	CL
CONF	CONF 3 >
AIR COND	On (STD) >

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Aircraft Status Panel

The area of the AIRCRAFT STATUS panel lists MEL and CDL items that have an impact on takeoff performance. If the aircraft is dispatched without any MEL/CDL item that affects takeoff performance, "MEL 0" and "CDL 0" appears in the MEL/CDL buttons labels.



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TAKEOFF RESULTS – OUTPUTS

1. GENERAL

When the runway selection and takeoff conditions are set, tap on “COMPUTE” to launch the computation.

The results panel is displayed when the computation is complete.

There are different possible types of output:

- Single results
- Multiple results.

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Example of main result page:



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Example of FMD-like result page:



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Example of AECG result page:



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Example of AECG result page:



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SINGLE RUNWAY COMPUTATION

Depending on the computation results, two cases are possible:

- Actual TOW above MTOW(PERF)
- Actual TOW equal to or below MTOW(PERF).

MTOW(PERF) is the highest TOW value for which takeoff is possible. As a result MTOW(PERF) can be higher than the weight variant MTOW.

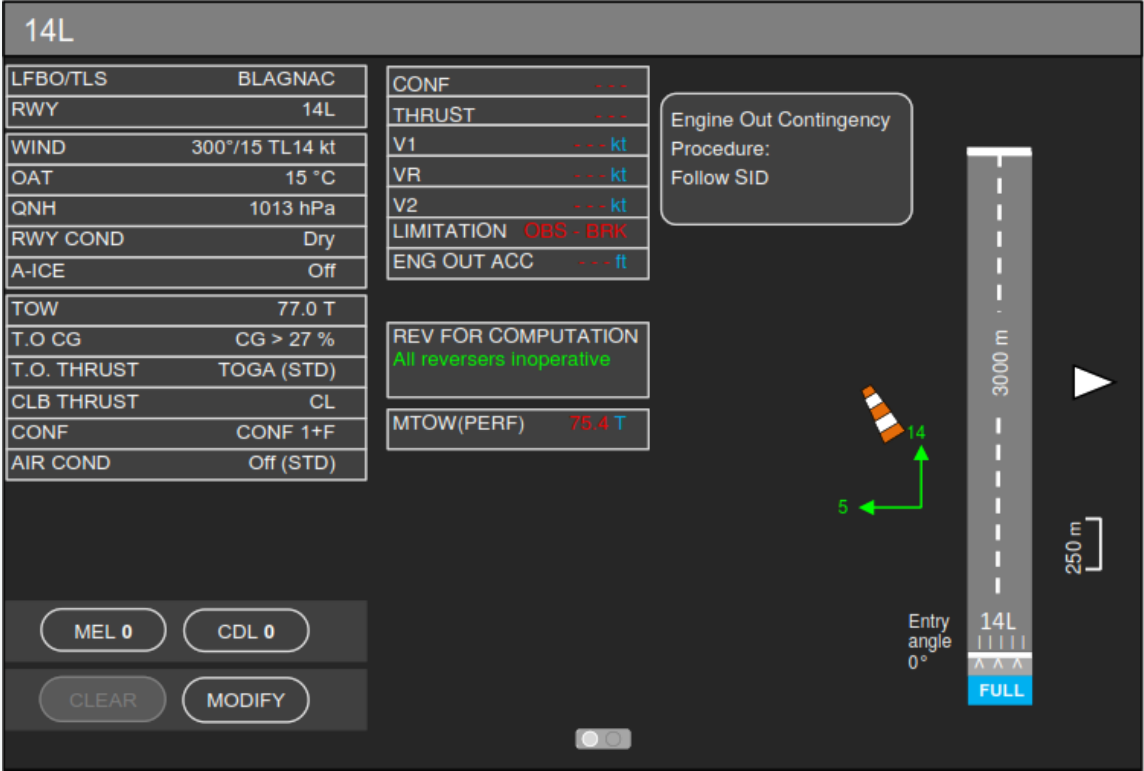
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Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW above MTOW (PERF)

When the calculated MTOW (perf) is below the TOW value (entered by the User), red dashes "----" are displayed in the results page.

The aircraft cannot be dispatched at this TOW on this runway. The takeoff weight has to be reduced to a value below the MTOW (perf) or a longer runway must be chosen, if any available on the considered airport.



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Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW equal to or below MTOW (PERF)

CONF

The Aircraft Configuration corresponds to the Configuration used for the computation. It can be:

- The configuration entered by the user in the CONDITION panel, if the user had chosen

CONF 1+F or CONF 2 or CONF 3

- The configuration that gives the best performance results if the user had chosen OPT CONF.

The screenshot displays the Airbus Flysmart App interface for runway computation. The top section shows the runway identifier '14L'. Below this, there are two main panels. The left panel lists various parameters: LFBO/TLS (BLAGNAC), RWY (14L), WIND (300°/15 TL14 kt), OAT (15 °C), QNH (1013 hPa), RWY COND (Dry), A-ICE (Off), TOW (77.0 T), T.O CG (CG > 27 %), T.O. THRUST (TOGA (STD)), CLB THRUST (CL), CONF (CONF 1+F), and AIR COND (Off (STD)). The right panel shows the configuration (CONF) and thrust (THRUST) settings, including V1, VR, V2, LIMITATION (OBS - BRK), and ENG OUT ACC. A box on the right indicates 'Engine Out Contingency Procedure: Follow SID'. Below these panels, there is a section for 'REV FOR COMPUTATION' with the note 'All reversers inoperative' and the MTOW(PERF) value of 75.4 T. At the bottom, there are buttons for MEL 0, CDL 0, CLEAR, and MODIFY. On the right side, there is a diagram of the runway layout showing the entry angle (0°) and the runway length (3000 m). The runway is labeled '14L' and 'FULL'.

14L	
LFBO/TLS	BLAGNAC
RWY	14L
WIND	300°/15 TL14 kt
OAT	15 °C
QNH	1013 hPa
RWY COND	Dry
A-ICE	Off
TOW	77.0 T
T.O CG	CG > 27 %
T.O. THRUST	TOGA (STD)
CLB THRUST	CL
CONF	CONF 1+F
AIR COND	Off (STD)

CONF	
THRUST	---
V1	--- kt
VR	--- kt
V2	--- kt
LIMITATION	OBS - BRK
ENG OUT ACC	--- ft

Engine Out Contingency Procedure: Follow SID

REV FOR COMPUTATION
All reversers inoperative

MTOW(PERF) 75.4 T

MEL 0 CDL 0

CLEAR MODIFY

Entry angle 0°

14L FULL

3000 m

250 m

Airbus Flysmart App

Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW equal to or below MTOW (PERF)

THRUST

The THRUST field displays the thrust setting used for the calculation.

- In case of takeoff at TOGA thrust, TOGA is displayed.
- In case of derated takeoff, it displays the derate level that can be either:
 - o The fixed derate level entered by the user in the THRUST field, or
 - o The optimum derate level if the user has selected the "OPTIMUM DERATE" option.

V1

V1 is the Takeoff decision speed. Only one V1 value is displayed and corresponds either to V1min, V1max or V1mean.

V1 RANGE

Any V1 value within the [V1 min; V1 max] range can be selected.

Airbus Flysmart App

Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW equal to or below MTOW (PERF)

THRUST

The THRUST field displays the thrust setting used for the calculation.

- In case of takeoff at TOGA thrust, TOGA is displayed.
- In case of derated takeoff, it displays the derate level that can be either:
 - o The fixed derate level entered by the user in the THRUST field, or
 - o The optimum derate level if the user has selected the "OPTIMUM DERATE" option.

V1

V1 is the Takeoff decision speed. Only one V1 value is displayed and corresponds either to V1min, V1max or V1mean.

Any V1 value within the [V1 min; V1 max] range can be selected. VR is the Rotation speed. V2 is the minimum climb speed that must be reached at a height of 35 ft above the runway surface, in case of an engine failure.

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Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW equal to or below MTOW (PERF)

Limitation code

Several regulatory requirements limit takeoff performance. They are called takeoff limitations. These takeoff limitations are identified in the computation results via limitation codes. The following table indicates the various codes that can appear:

TOW	Maximum computation weight, or maximum structural weight
RWY0	All-engine TakeOff Distance (TOD0), or All-engine TakeOff Run (TOR0), or All-engine Accelerate-Stop Distance (ASD0)
RWY1	One-engine inoperative TakeOff Distance (TOD1), or One-engine inoperative TakeOff Run (TOR1), or One-engine inoperative Accelerate-Stop Distance (ASD1)
VMCG	V1 limited by Minimum Control speed on Ground
TIRE	V _{LOF} limited by maximum TIRE speed (V _{LOF} is the speed at which the lift overcomes the weight)
BRK	V1 limited by maximum BRaKe energy
VMCA	V2 limited by Minimum Control speed in the Air
1SEG	1st SEGment climb gradient
2SEG	2nd SEGment climb gradient
3SEG	Acceleration (3 rd) segment
FTO	Final TakeOff climb gradient
OBS	OBStacle clearance
VMU	Minimum Unstick speed

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Airbus Takeoff application - SINGLE RUNWAY COMPUTATION

Actual TOW equal to or below MTOW (PERF)

Limitation code

ENG OUT ACC

The EO ACCEL field displays the suggested value of engine-out acceleration altitude. This value is a pressure altitude.

Green Dot Speed

The Green Dot field displays the value of Green Dot Speed.

REV for Computation

The field REV for Computation displays the number of reversers that are taken into account for the computation.

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Airbus Takeoff application - SINGLE RUNWAY COMPUTATION Actual TOW equal to or below MTOW (PERF)

Limitation code

Accelerate-Stop distance (ASD)

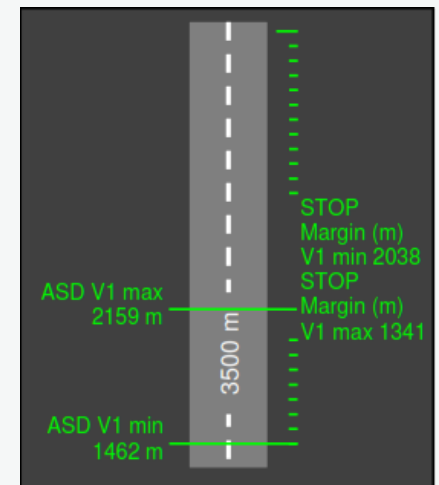
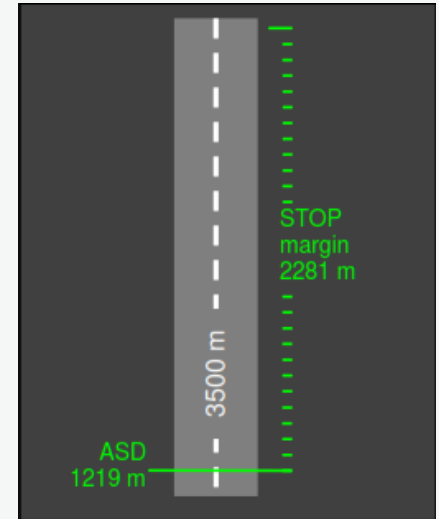
The ASD is displayed on the runway drawing. It corresponds to the distance necessary to perform a rejected takeoff.

Stop Margin

The stop margin is the difference between the ASD and the ASDA. It can be shown on the runway drawing, according to the administrator settings. Only one stop margin is displayed if the V1 type is:

- V1 min, or
- V1 mean, or
- V1 max.

Two stop margins are displayed if the V1 type is a range of possible V1 between V1min and V1max



Airbus Flysmart App

ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

The third results page that is related to the all engine gradient results contains two parts:

A reminder that displays the input value of the all engine gradient data A table with the all engine gradient results.

A "GRAPH" button is available in order to display the aircraft trajectory and/or to verify if the aircraft trajectory does not cross a published plan related to the constraint(s).

The horizontal axis is for the horizontal distance in nautical miles and the vertical axis is for the Altitude in feet.

We can differentiate two main cases, for which FlySmart+ does not display the same results First case is when there is no AECG constraint defined the runway And second case is when at least one AECG constraint is defined for the runway.

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ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

If no constraint is defined for the runway, FlySmart+ will provide the following information:

-The Minimum Average Climb Gradient (%), the Minimum Average Climb Gradient (ft/nm) and the Minimum Average Climb Speed (ft/min) are displayed in white color in the all engine gradient results page



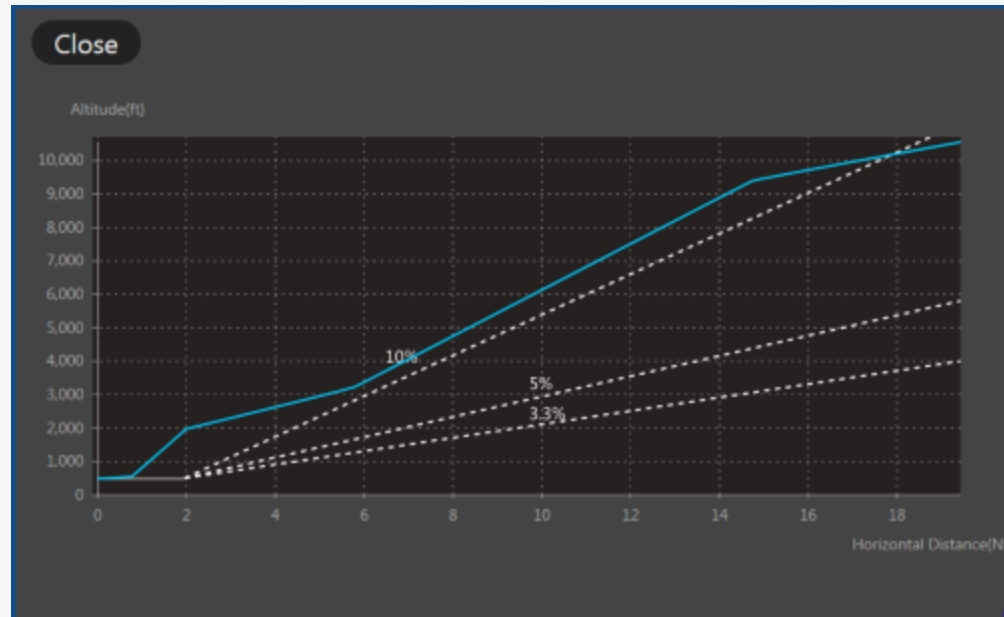
- The trajectory of the aircraft in blue color in the graphical results
- Plans related to a gradient value of 3.3 %, 5% and 10% in the graph results in white dashes.

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ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

The graph with the aircraft trajectory is the following:



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ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

With AECG constraints If the aircraft trajectory does not cross the published plan related to the constraint(s) AND if an average gradient to target altitude is above the related gradient of the constraint(s) that is defined:

FlySmart+ will provide the following information:

- The average gradient to the defined target altitude (% or ft/nm), and the average climb speed to the defined target altitude (ft/min) are displayed in green color in the all engine gradient results page
- The trajectory of the aircraft in blue color in the graphical results
- Published plans in white dashes lines, and if needed, the published plans are completed with the minimum regulatory gradient (3.3%) from the target altitude.

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ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

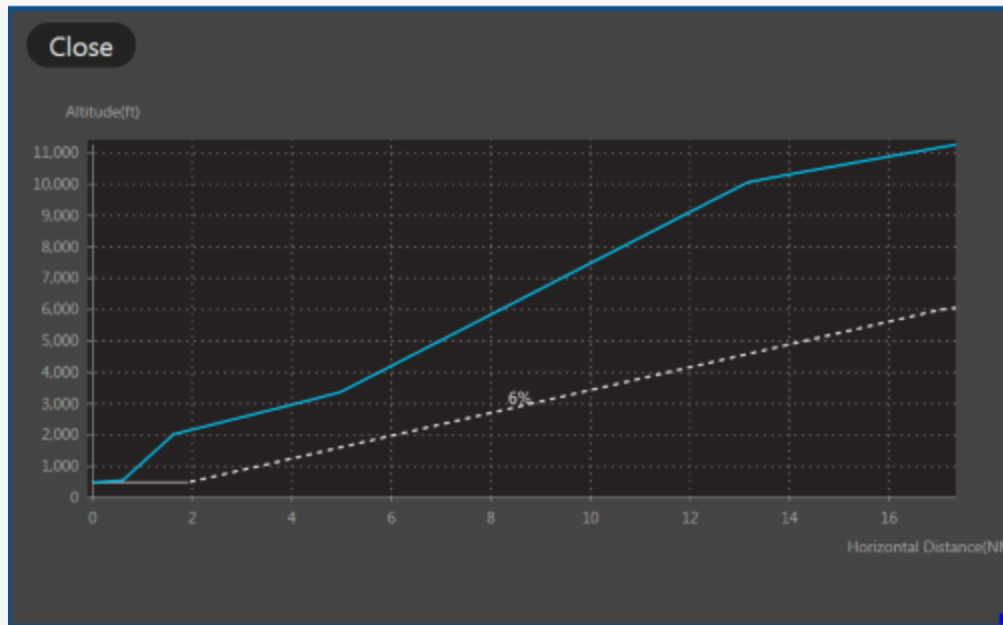


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ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

The graph with the aircraft trajectory is the following:



Airbus Flysmart App

ALL ENGINE CLIMB GRADIENT (AECG) OUTPUTS

No AECG constraint

If the aircraft trajectory crosses the published plan related to the constraint(s), OR if an average gradient to target altitude is below the defined constraint gradient to the target altitude:

FlySmart+ will provide the following information:

- A warning message "CHECK ALL ENG GRADIENT" that is displayed in amber and in capital letters on the first results page
- The outputs of AECG that lead to cross the published plan or the output gradient that is below to the defined constraint gradient are displayed in amber color in the all engine gradient results page
- The trajectory of the aircraft in blue color in the graphical results
- Published plans in white dashes lines, and if needed, the published plans are completed with the minimum regulation gradient (3.3%) from the target altitude.

Thank you.

